
Database Management Systems
IT354

Project

Deadline: Wednesday 3/12/2025 @ 23:59

[Total Marks for this Assignment is 14]

Student Details:

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Instructions:

- You must submit two separate copies (**one Word file and one PDF file**) using the Assignment Template on Blackboard via the allocated folder. These files **must not be in compressed format**.
- It is your responsibility to check and make sure that you have uploaded both the correct files.
- Zero mark will be given if you try to bypass the SafeAssign (e.g. misspell words, remove spaces between words, hide characters, use different character sets, **convert text into image** or languages other than English or any kind of manipulation).
- Email submission will not be accepted.
- You are advised to make your work clear and well-presented. This includes filling your information on the cover page.
- You must use this template, failing which will result in zero mark.
- You **MUST** show all your work, and text must not be converted into an image, unless specified otherwise by the question.
- Late submission will result in ZERO mark.
- The work should be your own, copying from students or other resources will result in ZERO mark.
- Use **Times New Roman** font for all your answers.

*Learning
Outcome(3):*

*Develop a
standard
database using
DBMS.*

Instructions

Imagine you are the database administrator for a large healthcare corporation, and you've been tasked with designing and implementing a new Clinic Management System database. This system addresses real-world needs such as organizing patient records, scheduling appointments, tracking treatments, and handling prescriptions efficiently.

- You must use **MySQL** for database creation and manipulation.
- **All** your answers should be according to the above scenario.
- Support your answer with a **detailed explanation** and **screenshots**.
- You should use the same project document to prepare your answer. **A word file and a pdf file should be provided.**
- Each group can have a **minimum of 2** and a **maximum of 4** students

*Learning
Outcome(4):*

*Analyze
algorithms for
query processing.*

Project Description

(14 Marks)

- **Part 1: Database Description and Design**

- 1- Write a brief introduction to a Clinic Management System database, including its purpose and at least five key entities. **(1 Mark)**
- 2- Create an Entity-Relationship Diagram (ERD). Ensure that cardinalities, relationships, and primary keys are clearly represented. State any assumptions that may affect the ER diagram. **(2 Marks)**
- 3- Convert the ER diagram to a class diagram. **(1 Mark)**

- **Part 2: Create and Populate Relations**

- 1- Map the ER diagram to a relational model and create the relations in MySQL **(1 Mark)**
- 2- Enter at least 15 tuples for each relation. **(1 Mark)**
Your solution should include screenshots of your relations with data and the query codes. Otherwise, marks will be deducted

- **Part 3: Indexes, Queries, and Fragmentation Type**

- 1- Write any retrieval query that includes a selection condition. Illustrate how MySQL internally performed this query (e.g., such as parsing and optimization) **(1 Mark)**
- 2- Create an index for the same column used in the previous selection condition. **(1 Mark)**
- 3- Choose a table from your database and apply vertical fragmentation to it. Present the table both before and after the fragmentation process. **(1 Mark)**
- 4- Write a query to retrieve data that includes only one join condition and one selection condition. Then, display the results of the query. **(1 Mark)**
- 5- Show the query statistics and execution plan for the above query. **(1 Mark)**

- **Part 4: Roles, Privileges and Triggers**

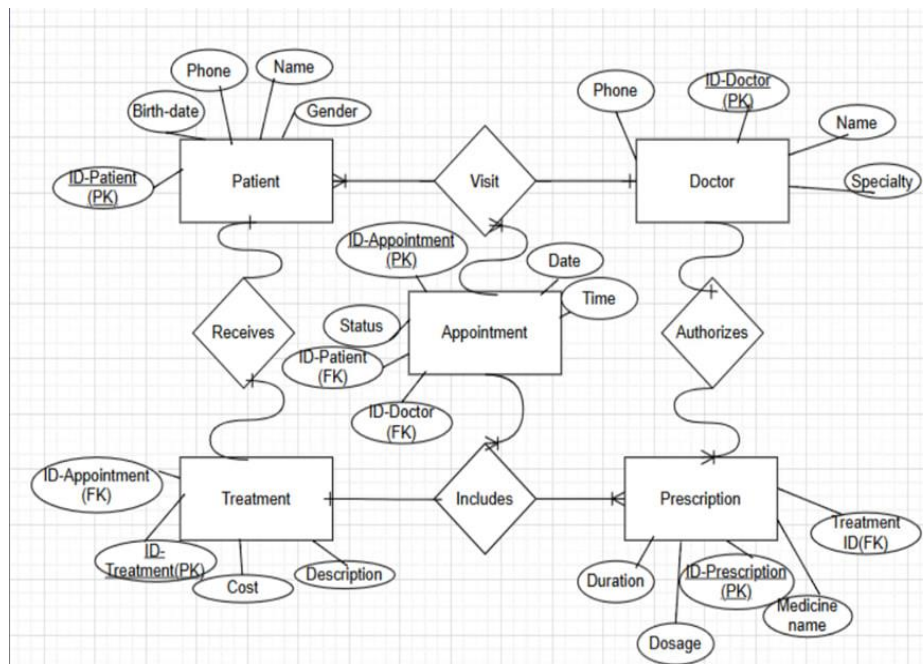
- 1- Create 4 roles and assign the following privileges to the roles: **(1 Mark)**
 - Give all privileges to the first role.
 - Give “insert” and “update” privileges to the second role, which can further assign the same privileges to others.
 - Give only select privileges to the third role, which can further assign the same privileges to others.
 - Give “update” privileges to the fourth role.
- 2- Create four accounts and assign the above roles to the created accounts (each account with a different role). **(1 Mark)**
- 3- Create an SQL trigger that verifies a specific condition in a table. **(1 Mark)**

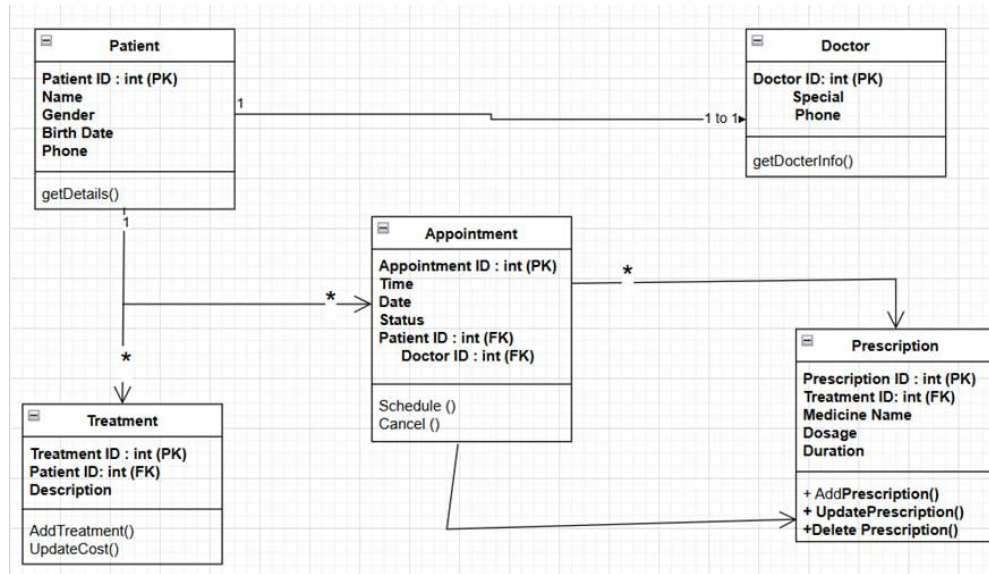
Part 1:

When we talk about clinics, we see that they face a crisis in organizing, retrieving, and storing patient data, doctor data, treatment data, prescription data, and many other problems related to data storage, access, and management. However, in this project, we will demonstrate how to solve these problems using a database that addresses all the aforementioned issues and more, transforming them into an easy-to-use tool that helps clinics improve their service and provide the best possible organization and management of information while maintaining complete confidentiality.

Five key entities: Patient- Doctor- Appointment- Treatment- Prescription

This will be achieved by using methods that facilitate access to, organize, and structure data, clarify appropriate relationships, define entities and attributes, and make it easy to access and use.





Part 2 of the project:

1) The Code:

```
CREATE DATABASE ClinicManagement;
USE ClinicManagement;
-- Create Doctors table
CREATE TABLE Doctors (
DoctorID INT PRIMARY KEY AUTO_INCREMENT,
DoctorName VARCHAR(50),
Specialty VARCHAR(50),
ExperienceYears INT
);
-- Insert 15 doctors
INSERT INTO Doctors (DoctorName, Specialty, ExperienceYears) VALUES
('Dr. Emad', 'Cardiology', 12),
('Dr. Zahraa', 'Dermatology', 8),
('Dr. Maryam', 'Pediatrics', 10),
('Dr. Abeer', 'Orthopedics', 7),
('Dr. Ahmad', 'Neurology', 15),
('Dr. Amal', 'Ophthalmology', 9),
('Dr. Tim', 'Dentistry', 11),
('Dr. Saeed', 'General Surgery', 14),
('Dr. Ayoub', 'Urology', 10),
('Dr. Majeed', 'ENT', 13),
('Dr. Sami', 'Psychiatry', 8),
('Dr. Shima', 'Gynecology', 9),
('Dr. Ohoud', 'Endocrinology', 10),
('Dr. Leen', 'Radiology', 6),
('Dr. Azeez', 'Emergency Medicine', 12);
-- Create Patients table
CREATE TABLE Patients (
PatientID INT PRIMARY KEY AUTO_INCREMENT,
PatientName VARCHAR(50),
Age INT,
NationalID VARCHAR(15)
);
-- Insert 15 patients
INSERT INTO Patients (PatientName, Age, NationalID) VALUES
('Layan', 22, '1023456789'),
('Samira', 25, '1029876543'),
('Rama', 20, '1034567891'),
```

```
('Najwa', 28, '1045678912'),
('Sara', 24, '1056789123'),
('Ebtehal', 26, '1067891234'),
('Areej', 21, '1078912345'),
('Jannah', 19, '1089123456'),
('Seham', 23, '1091234567'),
('Omar', 30, '1102345678'),
('Samer', 27, '1113456789'),
('Raed', 29, '1124567890'),
('Mohammed', 32, '1135678901'),
('Ammar', 35, '1146789012'),
('Walaa', 31, '1157890123');
-- Create Appointments table
CREATE TABLE Appointments (
AppointmentID INT PRIMARY KEY AUTO_INCREMENT,
PatientID INT,
DoctorID INT,
AppointmentDate DATE,
FOREIGN KEY (PatientID) REFERENCES Patients(PatientID),
FOREIGN KEY (DoctorID) REFERENCES Doctors(DoctorID)
);
-- Insert 15 appointments
INSERT INTO Appointments (PatientID, DoctorID, AppointmentDate) VALUES
(1, 3, '2025-11-01'),
(2, 5, '2025-11-02'),
(3, 2, '2025-11-03'),
(4, 8, '2025-11-04'),
(5, 1, '2025-11-05'),
(6, 4, '2025-11-06'),
(7, 7, '2025-11-07'),
(8, 6, '2025-11-08'),
(9, 9, '2025-11-09'),
(10, 10, '2025-11-10'),
(11, 11, '2025-11-11'),
(12, 12, '2025-11-12'),
(13, 13, '2025-11-13'),
(14, 14, '2025-11-14'),
(15, 15, '2025-11-15');
SELECT
a.AppointmentID,
p.PatientName,
p.Age,
```

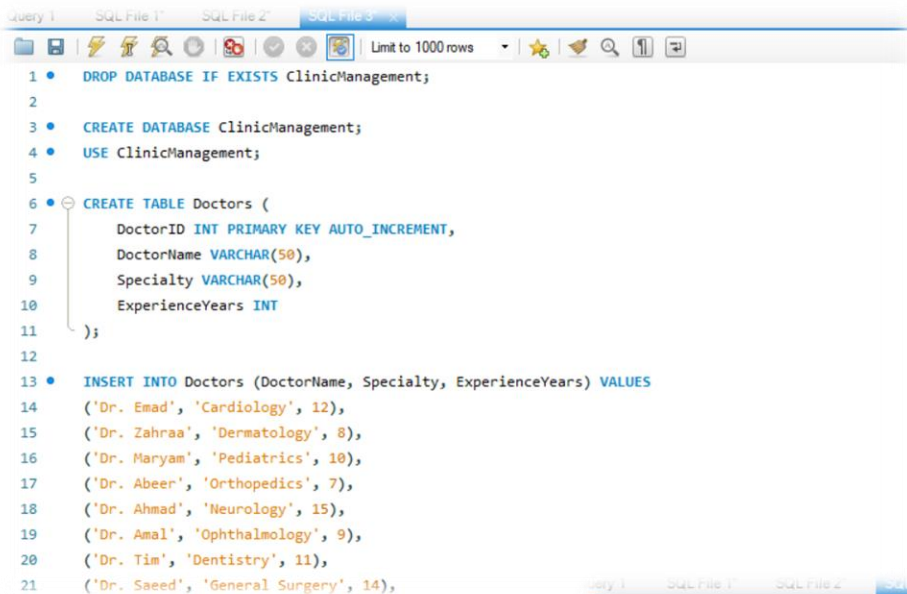


```

        p.NationalID,
        d.DoctorName,
        d.Specialty,
        d.ExperienceYears,
        a.AppointmentDate
    FROM Appointments a
    JOIN Patients p ON a.PatientID = p.PatientID
    JOIN Doctors d ON a.DoctorID = d.DoctorID;

```

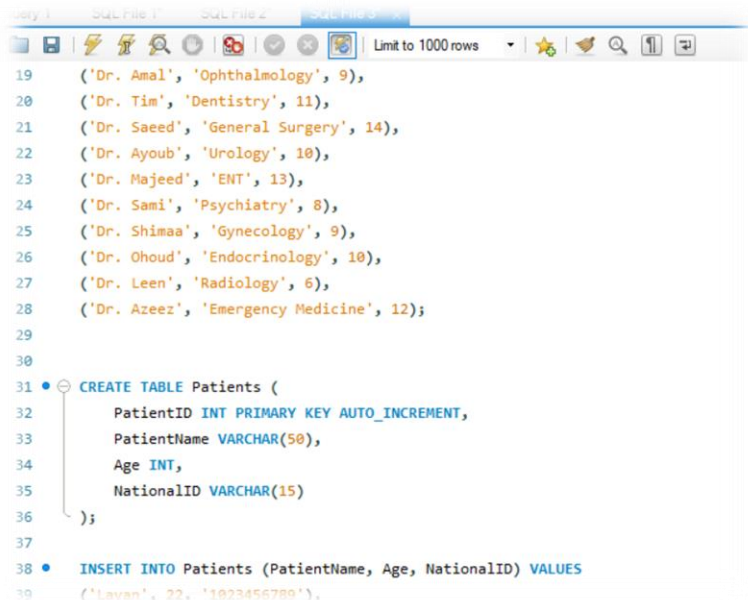
2) Screenshots of the code :



```

1  DROP DATABASE IF EXISTS ClinicManagement;
2
3  CREATE DATABASE ClinicManagement;
4  USE ClinicManagement;
5
6  CREATE TABLE Doctors (
7      DoctorID INT PRIMARY KEY AUTO_INCREMENT,
8      DoctorName VARCHAR(50),
9      Specialty VARCHAR(50),
10     ExperienceYears INT
11 );
12
13 INSERT INTO Doctors (DoctorName, Specialty, ExperienceYears) VALUES
14 ('Dr. Emad', 'Cardiology', 12),
15 ('Dr. Zahraa', 'Dermatology', 8),
16 ('Dr. Maryam', 'Pediatrics', 10),
17 ('Dr. Abeer', 'Orthopedics', 7),
18 ('Dr. Ahmad', 'Neurology', 15),
19 ('Dr. Amal', 'Ophthalmology', 9),
20 ('Dr. Tim', 'Dentistry', 11),
21 ('Dr. Saeed', 'General Surgery', 14),

```



```

19 ('Dr. Amal', 'Ophthalmology', 9),
20 ('Dr. Tim', 'Dentistry', 11),
21 ('Dr. Saeed', 'General Surgery', 14),
22 ('Dr. Ayoub', 'Urology', 10),
23 ('Dr. Majeed', 'ENT', 13),
24 ('Dr. Sami', 'Psychiatry', 8),
25 ('Dr. Shima', 'Gynecology', 9),
26 ('Dr. Ohoud', 'Endocrinology', 10),
27 ('Dr. Leen', 'Radiology', 6),
28 ('Dr. Azeez', 'Emergency Medicine', 12);
29
30
31 CREATE TABLE Patients (
32     PatientID INT PRIMARY KEY AUTO_INCREMENT,
33     PatientName VARCHAR(50),
34     Age INT,
35     NationalID VARCHAR(15)
36 );
37
38 INSERT INTO Patients (PatientName, Age, NationalID) VALUES
39 ('Layan', 22, '1023456789'),

```

```

60 FOREIGN KEY (PatientID) REFERENCES Patients(PatientID),
61 FOREIGN KEY (DoctorID) REFERENCES Doctors(DoctorID)
62 );
63
64
65 • INSERT INTO Appointments (PatientID, DoctorID, AppointmentDate) VALUES
66 (1, 3, '2025-11-01'),
67 (2, 5, '2025-11-02'),
68 (3, 2, '2025-11-03'),
69 (4, 8, '2025-11-04'),
70 (5, 1, '2025-11-05'),
71 (6, 4, '2025-11-06'),
72 (7, 7, '2025-11-07'),
73 (8, 6, '2025-11-08'),
74 (9, 9, '2025-11-09'),
75 (10, 10, '2025-11-10'),
76 (11, 11, '2025-11-11'),
77 (12, 12, '2025-11-12'),
78 (13, 13, '2025-11-13'),
79 (14, 14, '2025-11-14'),
80 (15, 15, '2025-11-15');

```

```

39 ('Layan', 22, '1023456789'),
40 ('Samera', 25, '1029876543'),
41 ('Rama', 20, '1034567891'),
42 ('Najwa', 28, '1045678912'),
43 ('Sara', 24, '1056789123'),
44 ('Ebtehal', 26, '1067891234'),
45 ('Areej', 21, '1078912345'),
46 ('Jannah', 19, '1089123456'),
47 ('Seham', 23, '1091234567'),
48 ('Omar', 30, '1102345678'),
49 ('Samer', 27, '1113456789'),
50 ('Raed', 29, '1124567890'),
51 ('Mohammed', 32, '1135678901'),
52 ('Ammar', 35, '1146789012'),
53 ('Walaa', 31, '1157890123');
54
55 • CREATE TABLE Appointments (
56 AppointmentID INT PRIMARY KEY AUTO_INCREMENT,
57 PatientID INT,
58 DoctorID INT,
59 AppointmentDate DATE,
60 FOREIGN KEY (PatientID) REFERENCES Patients(PatientID),
61 FOREIGN KEY (DoctorID) REFERENCES Doctors(DoctorID)
62 );

```

```
--
81
82
83 • SELECT
84     a.AppointmentID,
85     p.PatientName,
86     p.Age,
87     p.NationalID,
88     d.DoctorName,
89     d.Specialty,
90     d.ExperienceYears,
91     a.AppointmentDate
92 FROM Appointments a
93 JOIN Patients p ON a.PatientID = p.PatientID
94 JOIN Doctors d ON a.DoctorID = d.DoctorID
95 ORDER BY a.AppointmentID;
```

3) Screenshot of the output:

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	AppointmentID	PatientName	Age	NationalID	DoctorName	Specialty	ExperienceYears	AppointmentDate
▶	1	Layan	22	1023456789	Dr. Maryam	Pediatrics	10	2025-11-01
	2	Samera	25	1029876543	Dr. Ahmad	Neurology	15	2025-11-02
	3	Rama	20	1034567891	Dr. Zahraa	Dermatology	8	2025-11-03
	4	Najwa	28	1045678912	Dr. Saeed	General Surgery	14	2025-11-04
	5	Sara	24	1056789123	Dr. Emad	Cardiology	12	2025-11-05
	6	Ebtehal	26	1067891234	Dr. Abeer	Orthopedics	7	2025-11-06
	7	Areej	21	1078912345	Dr. Tim	Dentistry	11	2025-11-07
	8	Jannah	19	1089123456	Dr. Amal	Ophthalmology	9	2025-11-08
	9	Seham	23	1091234567	Dr. Ayoub	Urology	10	2025-11-09
	10	Omar	30	1102345678	Dr. Majeed	ENT	13	2025-11-10
	11	Samer	27	1113456789	Dr. Sami	Psychiatry	8	2025-11-11
	12	Raed	29	1124567890	Dr. Shimaa	Gynecology	9	2025-11-12
	13	Mohammed	32	1135678901	Dr. Ohoud	Endocrinology	10	2025-11-13
	14	Ammar	35	1146789012	Dr. Leen	Radiology	6	2025-11-14
	15	Walaa	31	1157890123	Dr. Azeez	Emergency Me...	12	2025-11-15

Part 3:**1. Retrieval Query with a Selection Condition and How MySQL Processes It**

A suitable retrieval query from the database is:

```
SELECT DoctorName, Specialty  
FROM Doctors  
WHERE ExperienceYears > 10;
```

```
77  
78 SELECT DoctorName, Specialty  
79 FROM Doctors  
80 WHERE ExperienceYears > 10;  
81  
82
```

≡ line 82, column 1, location 2268

	DoctorName	Specialty	
1	Dr. Emad	Cardiology	
2	Dr. Ahmad	Neurology	
3	Dr. Tim	Dentistry	
4	Dr. Saeed	General Surgery	
5	Dr. Majeed	ENT	
6	Dr. Azeez	Emergency Medicine	

Data Message Chart 2 ms

When this query is executed, MySQL first parses the statement to check its syntax and ensure all referenced columns and tables exist. After parsing, the optimizer evaluates possible execution strategies, such as whether to perform a full table scan or apply an

index. Since no index exists yet on ExperienceYears, the optimizer selects a full scan, comparing each row's ExperienceYears value against the condition. Finally, the execution engine retrieves only the rows that meet the selection criterion and returns them to the user.

2. Creating an Index on the Selection Column

To improve the performance of the previous query, an index can be created on the ExperienceYears column:

CREATE INDEX idx_experience ON Doctors(ExperienceYears);

After creating this index, MySQL can locate the qualified rows more efficiently by referencing the indexed values rather than scanning the entire table.

```
77
78 SELECT DoctorName, Specialty
79 FROM Doctors |
80 WHERE ExperienceYears > 10;
81
82 CREATE INDEX idx_experience ON Doctors(ExperienceYears);
83
84
```

line 79, column 14, location 2237

	DoctorName	Specialty	
1	Dr. Tim	Dentistry	
2	Dr. Emad	Cardiology	
3	Dr. Azeez	Emergency Medicine	
4	Dr. Majeed	ENT	
5	Dr. Saeed	General Surgery	
6	Dr. Ahmad	Neurology	

Data Message Chart 1 ms

3. Vertical Fragmentation of a Table

For this requirement, the Patients table is selected.

Before Vertical Fragmentation (Original Table):

Patients(PatientID, PatientName, Age, NationalID)

After Vertical Fragmentation:

Fragment 1:

Patients_Basic(PatientID, PatientName)

Fragment 2:

Patients_Details(PatientID, Age, NationalID)

The primary key PatientID remains in both fragments to maintain the ability to reconstruct the original relation when needed.

4. Query with One Join Condition and One Selection Condition

A query that retrieves appointment details with both a join and a selection condition is:

```
SELECT Patients.PatientName, Appointments.AppointmentDate  
FROM Appointments  
JOIN Patients ON Appointments.PatientID = Patients.PatientID  
WHERE Appointments.DoctorID = 3;
```

83

84 SELECT Patients.PatientName, Appointments.AppointmentDate

85 FROM Appointments

86 JOIN Patients ON Appointments.PatientID = Patients.PatientID

87 WHERE Appointments.DoctorID = 3;

88

89

90

line 84, column 28, location 2353

	PatientName	AppointmentDate	
1	Layan	2025-11-01	↕

5. Query Statistics and Execution Plan

The execution plan for the above query can be obtained using:

```
EXPLAIN
SELECT Patients.PatientName, Doctors.DoctorName,
Appointments.AppointmentDate
FROM Appointments
JOIN Patients ON Appointments.PatientID = Patients.PatientID
WHERE Appointments.DoctorID = 3;
```

90 EXPLAIN

91 SELECT Patients.PatientName, Appointments.AppointmentDate

92 FROM Appointments

93 JOIN Patients ON Appointments.PatientID = Patients.PatientID

94 WHERE Appointments.DoctorID = 3;

95

96

line 91, column 29, location 2541

	id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	1	SIMPLE	Appointments	NULL	ref	PatientID,DoctorID	DoctorID	5	const	1	100.00	Using where
2	1	SIMPLE	Patients	NULL	eq_ref	PRIMARY	PRIMARY	4	it354_project.Appointments.PatientID	1	100.00	NULL

The plan typically shows the order in which tables are accessed, the indexes MySQL chooses to apply, join type, estimated row counts, and filtering percentages. With no index on DoctorID, MySQL performs a full scan of the Appointments table, then uses a

primary key lookup for matching Patients rows. If an index existed on DoctorID, the plan would indicate index usage and a reduced number of examined rows.

Part 4: CODE:

```
DROP ROLE IF EXISTS role_admin;
DROP ROLE IF EXISTS role_editor;
DROP ROLE IF EXISTS role_reader;
DROP ROLE IF EXISTS role_updater;
DROP USER IF EXISTS 'hms_admin'@'localhost';
DROP USER IF EXISTS 'layan_editor'@'localhost';
DROP USER IF EXISTS 'areej_reader'@'localhost';
DROP USER IF EXISTS 'jannah_updater'@'localhost';
USE clinicmanagement;

CREATE ROLE role_admin;
GRANT ALL PRIVILEGES ON clinicmanagement.* TO role_admin;
CREATE ROLE role_editor;
GRANT INSERT, UPDATE ON clinicmanagement.* TO role_editor;
CREATE ROLE role_reader;
GRANT SELECT ON clinicmanagement.* TO role_reader WITH GRANT OPTION;
CREATE ROLE role_updater;
GRANT UPDATE ON clinicmanagement.* TO role_updater WITH GRANT OPTION;

CREATE USER 'hms_admin'@'localhost' IDENTIFIED BY 'hms2345';
CREATE USER 'layan_editor'@'localhost' IDENTIFIED BY 'layan5678';
CREATE USER 'areej_reader'@'localhost' IDENTIFIED BY 'areej9012';
CREATE USER 'jannah_updater'@'localhost' IDENTIFIED BY 'jannah8923';

GRANT role_admin TO 'hms_admin'@'localhost';
GRANT role_editor TO 'layan_editor'@'localhost';
GRANT role_reader TO 'areej_reader'@'localhost';
GRANT role_updater TO 'jannah_updater'@'localhost';

SET DEFAULT ROLE ALL TO 'hms_admin'@'localhost';
SET DEFAULT ROLE ALL TO 'layan_editor'@'localhost';
SET DEFAULT ROLE ALL TO 'areej_reader'@'localhost';
SET DEFAULT ROLE ALL TO 'jannah_updater'@'localhost';

DELIMITER $$
CREATE TRIGGER validate_patient_data
BEFORE INSERT ON Patients
FOR EACH ROW
```

```
BEGIN
    IF NEW.Age <= 0 OR NEW.Age > 120 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Age must be between 1 and 120';
    END IF;

    IF NEW.NationalID IS NULL OR NEW.NationalID = " THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'NationalID cannot be empty';
    END IF;
END$$
DELIMITER ;

SHOW TRIGGERS;

INSERT INTO Patients (PatientName, Age, NationalID) VALUES ('Rawan', 30, '2399871504');
INSERT INTO Patients (PatientName, Age, NationalID) VALUES ('Manal', 0, '1132980756');
INSERT INTO Patients (PatientName, Age, NationalID) VALUES ('Yara', 25, "");

SELECT * FROM Patients ORDER BY PatientID DESC LIMIT 5;

SELECT User, Host FROM mysql.user
WHERE User IN ('hms_admin', 'layan_editor', 'areej_reader', 'jannah_updater');

SHOW GRANTS FOR 'hms_admin'@'localhost';
SHOW GRANTS FOR 'layan_editor'@'localhost';
SHOW GRANTS FOR 'areej_reader'@'localhost';
SHOW GRANTS FOR 'jannah_updater'@'localhost';
```

Screenshot of the code:

```
1 • DROP ROLE IF EXISTS role_admin;
2 • DROP ROLE IF EXISTS role_editor;
3 • DROP ROLE IF EXISTS role_reader;
4 • DROP ROLE IF EXISTS role_updater;
5 • DROP USER IF EXISTS 'hms_admin'@'localhost';
6 • DROP USER IF EXISTS 'layan_editor'@'localhost';
7 • DROP USER IF EXISTS 'areej_reader'@'localhost';
8 • DROP USER IF EXISTS 'jannah_updater'@'localhost';
9 • USE clinicmanagement;
10
11 • CREATE ROLE role_admin;
12 • GRANT ALL PRIVILEGES ON clinicmanagement.* TO role_admin;
13 • CREATE ROLE role_editor;
14 • GRANT INSERT, UPDATE ON clinicmanagement.* TO role_editor;
15 • CREATE ROLE role_reader;
16 • GRANT SELECT ON clinicmanagement.* TO role_reader;
17 • CREATE ROLE role_updater;
18 • GRANT UPDATE ON clinicmanagement.* TO role_updater;
19
20 • CREATE USER 'hms_admin'@'localhost' IDENTIFIED BY 'hms2345';
21 • CREATE USER 'layan_editor'@'localhost' IDENTIFIED BY 'layan5678';
22 • CREATE USER 'areej_reader'@'localhost' IDENTIFIED BY 'areej9012';
23 • CREATE USER 'jannah_updater'@'localhost' IDENTIFIED BY 'jannah8923';
24
25 • GRANT role_admin TO 'hms_admin'@'localhost';
26 • GRANT role_editor TO 'layan_editor'@'localhost';

20 • CREATE USER 'hms_admin'@'localhost' IDENTIFIED BY 'hms2345';
21 • CREATE USER 'layan_editor'@'localhost' IDENTIFIED BY 'layan5678';
22 • CREATE USER 'areej_reader'@'localhost' IDENTIFIED BY 'areej9012';
23 • CREATE USER 'jannah_updater'@'localhost' IDENTIFIED BY 'jannah8923';
24
25 • GRANT role_admin TO 'hms_admin'@'localhost';
26 • GRANT role_editor TO 'layan_editor'@'localhost';
27 • GRANT role_reader TO 'areej_reader'@'localhost';
28 • GRANT role_updater TO 'jannah_updater'@'localhost';
29
30 • SET DEFAULT ROLE ALL TO 'hms_admin'@'localhost';
31 • SET DEFAULT ROLE ALL TO 'layan_editor'@'localhost';
32 • SET DEFAULT ROLE ALL TO 'areej_reader'@'localhost';
33 • SET DEFAULT ROLE ALL TO 'jannah_updater'@'localhost';
```

```

32 • SET DEFAULT ROLE ALL TO 'areej_reader'@'localhost';
33 • SET DEFAULT ROLE ALL TO 'jannah_updater'@'localhost';
34
35 DELIMITER $$
36 • CREATE TRIGGER validate_patient_data
37 BEFORE INSERT ON Patients
38 FOR EACH ROW
39 BEGIN
40 IF NEW.Age <= 0 OR NEW.Age > 120 THEN
41 SIGNAL SQLSTATE '45000'
42 SET MESSAGE_TEXT = 'Age must be between 1 and 120';
43 END IF;
44
45 IF NEW.NationalID IS NULL OR NEW.NationalID = '' THEN
46 SIGNAL SQLSTATE '45000'
47 SET MESSAGE_TEXT = 'NationalID cannot be empty';
48 END IF;
49 END$$
50 DELIMITER ;
51
52 • SHOW TRIGGERS;
53 • INSERT INTO Patients (PatientName, Age, NationalID) VALUES ('Rawan', 30, '2399871504');
54 • INSERT INTO Patients (PatientName, Age, NationalID) VALUES ('Manal', 0, '1132980756');
55 • INSERT INTO Patients (PatientName, Age, NationalID) VALUES ('Yara', 25, '');
56
57 • SELECT * FROM Patients ORDER BY PatientID DESC LIMIT 5;
58

```

```

--
52 • SHOW TRIGGERS;
53 • INSERT INTO Patients (PatientName, Age, NationalID) VALUES ('Rawan', 30, '2399871504');
54 • INSERT INTO Patients (PatientName, Age, NationalID) VALUES ('Manal', 0, '1132980756');
55 • INSERT INTO Patients (PatientName, Age, NationalID) VALUES ('Yara', 25, '');
56
57 • SELECT * FROM Patients ORDER BY PatientID DESC LIMIT 5;
58
59 • SELECT User, Host FROM mysql.user
60 WHERE User IN ('hms_admin', 'layan_editor', 'areej_reader', 'jannah_updater');
61
62 • SHOW GRANTS FOR 'hms_admin'@'localhost';
63 • SHOW GRANTS FOR 'layan_editor'@'localhost';
64 • SHOW GRANTS FOR 'areej_reader'@'localhost';
65 • SHOW GRANTS FOR 'jannah_updater'@'localhost';

```

OUT PUT OF THE CODE:

- TRIGGERS:

✓	37	23:51:34	CREATE TRIGGER validate_patient_data BEFORE INSERT ON Patients FOR EACH ROW BEGIN IF ...	0 row(s) affected
✓	38	23:53:09	INSERT INTO Patients (PatientName, Age, NationalID) VALUES ('Rawan', 30, '2399871504')	1 row(s) affected
✗	39	23:53:38	INSERT INTO Patients (PatientName, Age, NationalID) VALUES ('Manal', 0, '1132980756')	Error Code: 1644. Age must be between 1 and 120
✗	40	23:54:02	INSERT INTO Patients (PatientName, Age, NationalID) VALUES ('Yara', 25, '')	Error Code: 1644. NationalID cannot be empty
✓	41	23:56:20	SELECT * FROM Patients ORDER BY PatientID DESC LIMIT 5	5 row(s) returned

- SELECT * FROM Patients ORDER BY PatientID DESC LIMIT 5;

Result Grid

Filter Rows:

Edit:

	PatientID	PatientName	Age	NationalID
▶	20	Rawan	30	2399871504
	19	Yara	25	
	18	Yara	25	
	17	Manal	0	1132980756
	16	Rawan	30	2399871504
•	NULL	NULL	NULL	NULL

Patients 3

- SELECT User, Host FROM mysql.user

WHERE User IN ('hms_admin', 'layan_editor', 'areej_reader', 'jannah_updater');

Result Grid			Filter Rows:	
	User	Host		
▶	areej_reader	localhost		
	hms_admin	localhost		
	jannah_updater	localhost		
	layan_editor	localhost		
•	NULL	NULL		

SHOW GRANT :

	Grants for hms_admin@localhost
▶	GRANT USAGE ON *.* TO `hms_admin` @`localhost`
	GRANT `role_admin` @`%` TO `hms_admin` @`localhost`

	Grants for layan_editor@localhost
▶	GRANT USAGE ON *.* TO `layan_editor` @`localhost`
	GRANT `role_editor` @`%` TO `layan_editor` @`localhost`

	Grants for areej_reader@localhost
▶	GRANT USAGE ON *.* TO `areej_reader` @`localhost`
	GRANT `role_reader` @`%` TO `areej_reader` @`localhost`

	Grants for jannah_updater@localhost
▶	GRANT USAGE ON *.* TO `jannah_updater` @`localhost`
	GRANT `role_updater` @`%` TO `jannah_updater` @`localhost`